

主题(Topics)

- 发布主题（向设备发布指令）

Query/Request/{device_type}/{device_mn}

用途: 服务器向指定设备发送指令

{device_type}: 设备类型 (如: Tsp)

{device_mn}: 设备 mn (如: qqqqqq12345678)

- 订阅主题（接收设备响应）

Query/Response/{device_type}/{device_mn}

用途: 接收指定设备发送的响应数据

{device_type}: 设备类型 (如: Tsp)

{device_mn}: 设备 mn (如: qqqqqq12345678)

- 消息等级

Qos: 0

- 数据类型

Type: JSON/String

- 结束符

数据以 & 为结束符



数据格式规范

- 获取数据 (get_data)

◆ 请求示例 :

➤ Topic to publish: Query/Request/Tsp/qqqqqq12345678

➤ QoS: 0

➤ Payload Type: JSON/String

```
{
  "operation": "get_data",
```

```
"ids": [
  "a21004", "a21005", "a21026", "a21008",
  "a34005", "a34004", "a34002", "a34001",
  "a01001", "a01006", "a01007", "a01008",
  "a01002", "LA"
]
}&
```

◆ 响应成功示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "get_data",
  "code": "OK",
  "params": [
    "a21004", "a21005", "a21026", "a21008",
    "a34005", "a34004", "a34002", "a34001",
    "a01001", "a01006", "a01007", "a01008",
    "a01002", "LA"
  ],
  "values": [
    0.00, 0.00, 0.00, 0.00,
    0.00, 0.00, 0.00, 0.00,
    0.00, 0.00, 0.00, 0.00,
    0.00, 0.00
  ],
  "message": "get data success"
}&
```

◆ 响应失败示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "get_data",
  "code": "NG",
  "params": "NULL",
  "values": "NULL",
  "message": "error message"
}
```

● 设置 HJ212 协议，部分设备配置 (set_data)

◆ 请求示例

- Topic to publish: Query/Request/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "set_data",
  "ids": [
    { "id": "PW", "value": "123456" },
    { "id": "MN", "value": "qqqqqq12345678" },
    { "id": "ST", "value": "31" },
    { "id": "FLAG", "value": "9" }
  ]
}&
```

◆ 响应成功示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "set_data",
  "code": "OK",
  "params": ["PW", "MN", "ST", "FLAG"],
  "values": ["123456", "qqqqqq12345678", "31", "9"],
  "message": "Set PW success Set MN success Set ST success Set FLAG
success"
}&
```

◆ 响应失败示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "set_data",
  "code": "NG",
  "params": "NULL",
  "values": "NULL",
  "message": "error message"
}
```

```
}&
```

● 校准数据 (gal_data)

◆ 请求示例

- Topic to publish: Query/Request/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "gal_data",
  "ids": [
    {
      "id": "TSP",
      "params": ["DATA", "RATIO"],
      "values": [1, 100]
    },
    {
      "id": "SO2",
      "params": ["DATA", "RATIO"],
      "values": [0, 0]
    }
  ]
}&
```

◆ 响应成功示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "gal_data",
  "code": "OK",
  "params": "NULL",
  "values": "NULL",
  "message": "SO2 updatecal ok - TSP updatecal ok"
}&
```

◆ 响应失败示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0

➤ Payload Type: JSON/String

```
{
  "operation": "gal_data",
  "code": "NG",
  "params": "NULL",
  "values": "NULL",
  "message": "error message"
}&
```

● 获取配置 (get_config)

◆ 请求示例

- Topic to publish: Query/Request/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "get_config"
}&
```

◆ 响应成功示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "get_config",
  "code": "OK",
  "params": "config.JSON/String",
  "values": {
    "ip": "",
    "pw": "123456",
    "mn": "qqqqqq12345678",
    "flag": "9",
    "st": "31",
    "collect_ids": [
      "a21004", "a21005", "a21026", "a21008",
      "a34005", "a34004", "a34002", "a34001",
      "a01001", "a01006", "a01007", "a01008",
      "a01002", "LA"
    ]
  }
}
```

```
    },
    "message": "get config success"
  }&
```

◆ 响应失败示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "get_config",
  "code": "NG",
  "params": "NULL",
  "values": "NULL",
  "message": "error message"
}&
```

● 设置配置 (set_config)

◆ 请求示例

- Topic to publish: Query/Request/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "set_config",
  "config": {
    "ip": "39.101.67.255:8002",
    "mn": "testttt123456789",
    "pw": "123456",
    "st": "31",
    "flag": "9",
    "collect_ids": [
      "a21004", "a21005", "a21026", "a21008",
      "a34005", "a34004", "a34002", "a34001",
      "a01007", "a01008", "a01001", "a01019",
      "a01018", "LA"
    ]
  }
}&
```

◆ 响应成功示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "set_config",
  "code": "OK",
  "params": "config.json",
  "values": {
    "ip": "39.101.67.255:8002",
    "pw": "123456",
    "mn": "testtt123456789",
    "flag": "9",
    "st": "31",
    "collect_ids": [
      "a21004", "a21005", "a21026", "a21008",
      "a34005", "a34004", "a34002", "a34001",
      "a01007", "a01008", "a01001", "a01019",
      "a01018", "LA"
    ]
  },
  "message": "set config success"
}&
```

◆ 响应失败示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "set_config",
  "code": "NG",
  "params": "NULL",
  "values": "NULL",
  "message": "error message"
}&
```

● 获取设置数据 (get_set_data)

◆ 请求示例

- Topic to publish: Query/Request/Tsp/qqqqqq12345678
- QoS: 0

➤ Payload Type: JSON/String

```
{
  "operation": "get_set_data",
  "ids": ["ST", "MN", "PW", "FLAG", "TIME", "IP"]
}&
```

◆ 响应成功示例

➤ Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678

➤ QoS: 0

➤ Payload Type: JSON/String

```
{
  "operation": "get_set_data",
  "code": "OK",
  "params": ["ST", "MN", "PW", "FLAG", "TIME", "IP"],
  "values": ["31", "testttt123456789", "123456", "9", "NULL",
"39.101.67.255:8002"],
  "message": "get data success"
}&
```

◆ 响应失败示例

➤ Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678

➤ QoS: 0

➤ Payload Type: JSON/String

```
{
  "operation": "get_set_data",
  "code": "NG",
  "params": "NULL",
  "values": "NULL",
  "message": "error message"
}&
```

● 设置日志 (set_log)

◆ 请求示例

➤ Topic to publish: Query/Request/Tsp/qqqqqq12345678

➤ QoS: 0

➤ Payload Type: JSON/String

```
{
  "operation": "set_log",
```

```
"config": {
  "level": "debug", //( "info", "error","warning")
  "uart": "dtu" //( "usb")
}&
}
```

◆ 响应成功示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "set_log",
  "code": "OK",
  "params": "NULL",
  "values": {
    "level": "debug",
    "uart": "dtu"
  },
  "message": "set log success"
}&
```

◆ 响应失败示例

- Topic to Subscribe: Query/Response/Tsp/qqqqqq12345678
- QoS: 0
- Payload Type: JSON/String

```
{
  "operation": "set_log",
  "code": "NG",
  "params": "NULL",
  "values": "NULL",
  "message": "error message"
}&
```